Sahaj Bhadja

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Summary

Full-Stack Software Engineer with 2+ years building **Python** REST APIs, React and Angular front ends, and **PostgreSQL**-backed applications with secure **authentication**, **unit testing**, and Jenkins **CI/CD** in Agile teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Designed normalized **PostgreSQL** and Azure SQL schemas with row-level security for a multi-tenant platform serving 500+ users, optimizing indexes and query patterns to cut dashboard latency from 1.2s to 350ms.
- Built a fault-tolerant async Python job system on **FastAPI** with explicit state transitions and failure isolation, processing 1K+ AI jobs and tripling throughput under concurrent load for high availability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Built a modular Flask backend with Blueprint-based MVC structure, Flask-Login authentication, and **REST APIs** behind a React dashboard, serving 150+ instructors and 10K+ daily grading requests on AWS RDS.
- Implemented concurrent batch processing with ThreadPoolExecutor and secure PDF, DOCX, and ZIP ingestion, persisting normalized results and usage logs while cutting feedback turnaround 60% under peak load.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Built reusable **object-oriented** modules with Singleton and Factory patterns for loosely coupled microservices integrating multiple data sources, improving API response times 30% and cutting technical debt 20%.
- Wrote 20+ **unit** and integration test suites with PyTest and Mockito, running them through Jenkins and GitHub pipelines nightly across Windows, Linux, and macOS during Agile Scrum sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight microservices including JWT **authentication** and **role-based access** services for a MERN platform, securing Azure SQL data with encryption and firewalls while synchronizing 2M+ records across locations.
- Automated **Jenkins CI/CD** with linting, Jest and Mocha tests, Docker packaging, and staged deployments, achieving 95%+ successful build validation with zero-downtime rolling production updates on Azure App Services.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Developed Flask **REST APIs** for booking, pricing, coupons, and payments with JWT and wallet-signature authentication, sustaining sub-200ms latency across 50K+ records for React Native mobile and React web clients.
- Implemented idempotent payment webhooks and Firebase transactions that prevented double bookings on shared time slots, reducing manual refund handling by about 70% and keeping payment status consistent.

Skills

Back-end & Testing: Python, Django, FastAPI, Flask, REST APIs, Microservices, OOP, Node.js, Java, Unit Testing, Debugging
Front-end: Angular.js, JavaScript, TypeScript, HTML, CSS, React.js, Next.js, TailwindCSS, Responsive Design, Agile/Scrum
Data & Security: PostgreSQL, MySQL, SQL Server, MongoDB, SQL, JWT, Role-Based Access Control, Row-Level Security, NLP
Tools & Cloud: Git, JIRA, Jenkins, GitHub Actions, CI/CD, Docker, Kubernetes, AWS (EC2, S3, RDS), PyTest, Jest, Express.js

Projects

HMIS - Hospital Management Information System Live | *TypeScript, MERN, TailwindCSS, ABDM APIs, Sentry*

- Built a responsive TailwindCSS and TypeScript **clinical workflow** platform with consent-based patient-record access, appointment dashboards, and Sentry monitoring, supporting 100+ daily appointments and reducing administrative workload 30%.

PiXMod Live | *Angular.js, Next.js, MongoDB, Stripe, CLIP, SAM*

- Built an AI image SaaS with a responsive Next.js interface, MongoDB image and **metadata storage**, CLIP search across 100K+ indexed embeddings, and Stripe subscriptions that raised premium conversion 18%.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, TailwindCSS, Plaid*

- Built a Next.js and TypeScript fintech platform with Plaid bank linking, a token-based firewall for **secure data access**, and Kubernetes CI/CD deployments, supporting 50K+ users and 1M+ transactions.